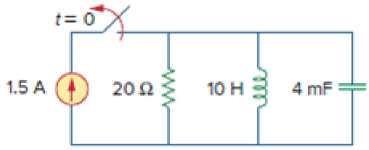
Chapter 8, Problem 6PP

Problem

Refer to the circuit in Fig. 8.17. Find $v(t)$ for $t > 0$.

Figure 8.17:



Step-by-step solution

Step 1 of 13

Refer to Figure 8.17 in the text book for the actual circuit.

At $t < 0$, the inductor acts as short circuit and the capacitor acts as open circuit.

Draw the circuit diagram at $t < 0$.

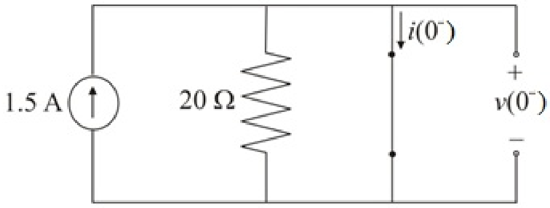


Figure 1

Step 2 of 13

Notice that the resistor is shorted. Hence, no current flows through the resistor. The current from the source flows through the shorted terminals of inductor. Therefore, the value of the current $i\left(0^-\right)$ is 1.5 A.

Since, the terminals of the capacitor are also shorted, the value of the voltage $v\left(0^-\right)$ is 0 V.

Step 3 of 13

The switch opens at $t>0$, hence circuit is independent of the source.

Draw the circuit diagram at $t>0$.

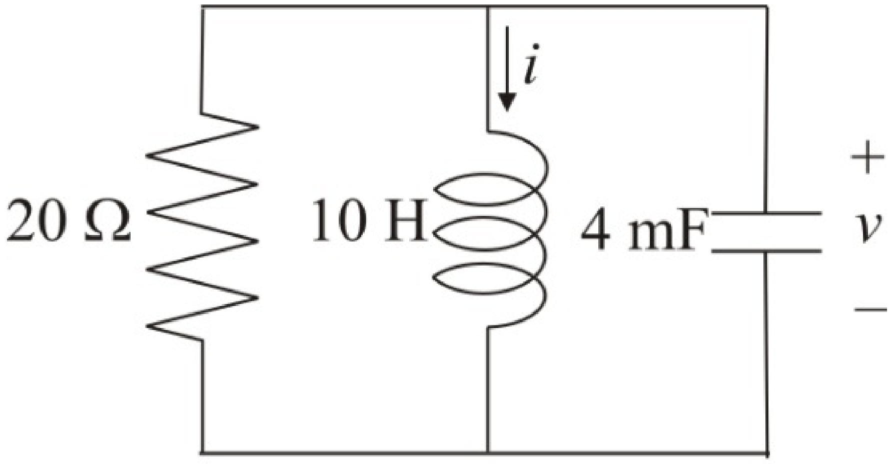


Figure 2

Step 4 of 13

The voltage across the capacitor and the current through the inductor doesn't change instantaneously. Hence, at $t>0$ the value of the voltage across the capacitor $v(0^+)$ is 0 V and the value of the current $i(0^+)$ is 1.5 A.

Notice that the voltage across the capacitor and the current through the inductor are continuous. Therefore, the values at $t=0$ are $v(0)=0\text{ V}$ and $i(0)=1.5\text{ A}$.

Step 5 of 13

Apply Kirchhoff's current law at top node of Figure 2.

$$\frac{v(t)}{R}+i(t)+C\frac{dv(t)}{dt}=0$$
$$C\frac{dv(t)}{dt}=-\frac{v(t)}{R}-i(t)$$
$$\frac{dv(t)}{dt}=-\frac{v(t)}{RC}-\frac{i(t)}{C}$$
$$\frac{dv(t)}{dt}=-\left(\frac{v(t)+Ri(t)}{RC}\right)$$

Substitute 0 for t in equation.

$$\begin{aligned}\frac{dv(0)}{dt}&=-\left(\frac{v(0)+Ri(0)}{RC}\right)\\&=-\left(\frac{0+(20)(1.5)}{(20)(4\times10^{-3})}\right)\\&=-375\text{ V/s}\end{aligned}$$

Therefore, the value of the $\dfrac{dv(0)}{dt}$ is -375 V/s.

Step 6 of 13

Calculate the value of the neper frequency α.

$$\alpha = \frac{1}{2RC}$$

Substitute $20\ \Omega$ for R and $4\ \text{mF}$ for C.

$$\begin{aligned}\alpha &= \frac{1}{2(20)(4\ \text{m})} \\ &= \frac{1}{160\ \text{m}} \\ &= 6.25\ \text{Np/s}\end{aligned}$$

Therefore, the value of the neper frequency α is $6.25\ \text{Np/s}$.

Step 7 of 13

Calculate the value of the resonant frequency ω_0.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute $10\ \text{H}$ for L and $4\ \text{mF}$ for C.

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{(10)(4\ \text{m})}} \\ &= \frac{1}{\sqrt{40\ \text{m}}} \\ &= \frac{1}{0.2} \\ &= 5\ \text{rad/s}\end{aligned}$$

Therefore, the value of the resonant frequency ω_0 is $5\ \text{rad/s}$.

Step 8 of 13

Consider the expression for the roots of the characteristic equation.

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

Substitute $5\ \text{rad/s}$ for ω_0 and $6.25\ \text{Np/s}$ for α.

$$\begin{aligned}s_{1,2} &= -6.25 \pm \sqrt{(6.25)^2 - (5)^2} \\ &= -6.25 \pm \sqrt{39.0625 - 25} \\ &= -6.25 \pm \sqrt{14.0625} \\ &= -6.25 \pm 3.75\end{aligned}$$

$s_1 = -6.25 - 3.75$ and $s_2 = -6.25 + 3.75$

$s_1 = -10$ and $s_2 = -2.5$

Therefore, the roots of the characteristic equation are -10 and -2.5.

Step 9 of 13

The value of the $\alpha > \omega_0$, hence the circuit exhibits over-damped response

Write the general solution for the voltage $v(t)$.

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Substitute -10 for s_1 and -2.5 for s_2.

$$v(t) = A_1 e^{-10t} + A_2 e^{-2.5t} \quad \text{...... (1)}$$

Step 10 of 13

Recall equation (1).

$$v(t) = A_1 e^{-10t} + A_2 e^{-2.5t}$$

Substitute 0 for t.

$$\begin{aligned} v(0) &= A_1 e^{-10(0)} + A_2 e^{-2.5(0)} \\ &= A_1 e^0 + A_2 e^0 \\ &= A_1(1) + A_2(1) \\ &= A_1 + A_2 \end{aligned}$$

Substitute 0 V for $v(0)$.

$$0 = A_1 + A_2$$

$$A_1 = -A_2 \quad \text{...... (2)}$$

Step 11 of 13

Recall equation (1).

$$v(t) = A_1 e^{-10t} + A_2 e^{-2.5t}$$

Apply derivative with respect to t.

$$\begin{aligned} \frac{dv(t)}{dt} &= A_1 \frac{d\left(e^{-10t}\right)}{dt} + A_2 \frac{d\left(e^{-2.5t}\right)}{dt} \\ &= -10A_1 e^{-10t} - 2.5A_2 e^{-2.5t} \end{aligned}$$

Substitute 0 for t.

$$\begin{aligned} \frac{dv(0)}{dt} &= -10A_1 e^{-10(0)} - 2.5A_2 e^{-2.5(0)} \\ &= -10A_1 e^{(0)} - 2.5A_2 e^{(0)} \\ &= -10A_1(1) - 2.5A_2(1) \\ &= -10A_1 - 2.5A_2 \end{aligned}$$

Substitute $-A_2$ for A_1 and -375 V/s for $\frac{dv(0)}{dt}$.

$$-375 = -10A_1 - 2.5A_2$$

$$7.5A_2 = -375$$

$$A_2 = \frac{-375}{7.5}$$

$$A_2 = -50$$

Therefore, the value of the constant A_2 is -50.

Step 12 of 13

Recall equation (2).

$$A_1 = -A_2$$

Substitute -50 for A_2.

$$A_1 = 50$$

Therefore, the value of the constant A_1 is 50.

Step 13 of 13

Recall equation (1).

$$v(t) = A_1 e^{-10t} + A_2 e^{-2.5t}$$

Substitute -50 for A_2 and 50 for A_1.

$$\begin{aligned} v(t) &= 50e^{-10t} - 50e^{-2.5t} \\ &= 50\left(e^{-10t} - e^{-2.5t}\right) \end{aligned}$$

Therefore, the value of the voltage $v(t)$ is $\boxed{50\left(e^{-10t} - e^{-2.5t}\right)\text{ V}}$.